

# Waves

## Question1

**When both source of sound and observer approach each other with a speed equal to 10% of the speed of sound, then the percentage change in frequency heard by the observer is nearly**

**AP EAPCET 2025 - 26th May Morning Shift**

**Options:**

A.

33.3%

B.

12.2%

C.

22.2%

D.

11.1%

**Answer: C**

**Solution:**

**Given:**

Both **source** and **observer** are moving **towards each other**.

Each moves at **10% of the speed of sound**.

$$v_s = 0.1v, \quad v_o = 0.1v$$

where  $v$  = speed of sound.

### Formula (for Doppler effect):

When both are moving **towards each other**, the apparent frequency  $f'$  heard by the observer is

$$f' = f \frac{v+v_o}{v-v_s}$$

### Substituting given values:

$$f' = f \frac{v+0.1v}{v-0.1v} = f \frac{1.1v}{0.9v} = f \times \frac{1.1}{0.9}$$

$$\frac{1.1}{0.9} = 1.222 \dots$$

So,

$$f' = 1.222f$$

### Percentage change in frequency:

$$\% \text{ increase} = (1.222 - 1) \times 100 = 22.2\%$$

 **Final Answer:**

22.2% (Option C)

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## Question2

**The equation of a transverse wave propagating along a stretched string of length 80 cm is  $y = 1.5 \sin \{ (5 \times 10^{-3}x) + 20t \}$ , here '  $x$  ' and '  $y$  ' are in cm and the time '  $t$  ' is in second. If the mass of the string is 3 g , then the tension in the string is 80 cm**

### AP EAPCET 2025 - 26th May Evening Shift

**Options:**

A.

12 N

B.

4 N

C.

6 N

D.

8 N

**Answer: C**

**Solution:**

From the given wave equation

$$k = 5 \times 10^{-3} \text{ cm}^{-1} = 5 \times 10^{-1} \text{ m}^{-1} \\ = 0.5 \text{ m}^{-1}$$

$$w = 20 \text{ rad/s}$$

$$\therefore v = \frac{\omega}{k} = \frac{20}{5 \times 10^{-1}} = 40 \text{ m/s}$$

$$\therefore \mu = \frac{m}{l} = \frac{3 \times 10^{-3}}{0.8} \\ = 3.75 \times 10^{-3} \text{ kg/m}$$

$$\therefore \text{Since, } v = \sqrt{\frac{T}{\mu}}$$

$$\Rightarrow v^2 = \frac{T}{\mu}$$

$$\Rightarrow T = \mu v^2 = 3.75 \times 10^{-3} \times 40^2 \\ = 3.75 \times 10^{-3} \text{ N} = 6 \text{ N}$$

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## Question3

**If a travelling wave is given by  $y(x, t) = 0.5 \sin(70.1x - 10\pi t)$ , where  $x$  and  $y$  are in metre the time  $t$  is in second, then the frequency of the wave is**

**AP EAPCET 2025 - 24th May Morning Shift**

**Options:**

A.

6 Hz

B.

7 Hz

C.

4 Hz

D.

5 Hz

**Answer: D**

### **Solution:**

Standard equation of a travelling wave is,

$$y = A \sin(kx - \omega t)$$

$$\text{Given, } y = 0.5 \sin(70.1x - 10\pi t)$$

$$\therefore \omega = \text{angular frequency} = 10\pi$$

$$\text{But } \omega = 2\pi f, f = \text{frequency}$$

$$\Rightarrow f = \frac{\omega}{2\pi} = \frac{10\pi}{2\pi} = 5 \text{ Hz}$$

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## **Question4**

**The path difference between two waves given by the equations**

$$y_1 = a_1 \sin\left(\omega t - \frac{2\pi x}{\lambda}\right) \text{ and } y_2 = a_2 \sin\left(\omega t - \frac{2\pi x}{\lambda} + \phi\right) \text{ is}$$

**AP EAPCET 2025 - 24th May Morning Shift**

**Options:**

A.

$$\left(\frac{\lambda}{\pi}\phi\right)$$

B.

$$\frac{\lambda}{\pi}\left(\phi - \frac{\pi}{2}\right)$$

C.

$$\frac{\lambda}{2\pi} \phi$$

D.

$$\frac{\lambda}{2\pi} \left( \phi - \frac{\pi}{2} \right)$$

**Answer: C**

**Solution:**

Phase difference between waves is

$$\begin{aligned} \phi_1 - \phi_2 &= \left( \omega t - \frac{2\pi x}{\lambda} + \phi \right) - \left( \omega t - \frac{2\pi x}{\lambda} \right) \\ &= \phi \end{aligned}$$

As, relation of phase and path difference is,

$$\frac{\Delta \phi}{2\pi} = \frac{\Delta L}{\lambda}$$

$$\Rightarrow \text{Path difference, } \Delta L = \frac{\lambda \phi}{2\pi}.$$

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## Question5

**If two progressive sound waves represented by  $y_1 = 3 \sin 250\pi t$  and  $y_2 = 2 \sin 260\pi t$  (where displacement is in metre and time is in second) superimpose, then the time interval between two successive maximum intensities is**

**AP EAPCET 2025 - 23rd May Evening Shift**

**Options:**

A.

0.1 s

B.

0.4 s

C.

0.5 s

D.

0.2 s

**Answer: D**

**Solution:**

$$\begin{aligned}y_1 &= 3 \sin 250\pi t \\ \therefore \omega_1 &= 250\pi \text{ rad/s} \\ y_2 &= 2 \sin 260\pi t \\ \therefore \omega_2 &= 260\pi \text{ rad/s} \\ \therefore f_1 &= \frac{\omega_1}{2\pi} = \frac{250\pi}{2\pi} = 125 \text{ Hz} \\ f_2 &= \frac{\omega_2}{2\pi} = \frac{260\pi}{2\pi} = 130 \text{ Hz} \\ T &= \frac{1}{f_2 - f_1} \\ &= \frac{1}{130 - 125} = \frac{1}{5} = 0.2 \text{ s}\end{aligned}$$

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## Question6

**In a closed organ pipe, the number of nodes formed in fifth and ninth harmonics are respectively**

**AP EAPCET 2025 - 23rd May Morning Shift**

**Options:**

A.

5,9

B.

5,7

C.

3,5

D.

2,4

**Answer: C**

**Solution:**

In a closed organ pipe, the number of nodes formed in the  $n$ th harmonic is equal to  $\left(\frac{n+1}{2}\right)$  if  $n$  is odd.

Thus, for the fifth ( $n = 5$ ) harmonic, the number of nodes =  $\frac{5+1}{2} = 3$

and for the ninth harmonic ( $n = 9$ ), the number of nodes =  $\frac{9+1}{2} = 5$

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## Question7

**When a stretched wire of fundamental frequency  $f$  is divided into three segments, the fundamental frequencies of these three segments are  $f_1$ ,  $f_2$  and  $f_3$  respectively. Then the relation among  $f_1$ ,  $f_2$ ,  $f_3$  and  $f$  is (Assume tension is constant)**

**AP EAPCET 2025 - 22nd May Evening Shift**

**Options:**

A.

$$\sqrt{f} = \sqrt{f_1} + \sqrt{f_2} + \sqrt{f_3}$$

B.

$$f = f_1 + f_2 + f_3$$

C.

$$\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}$$

D.

$$\frac{1}{\sqrt{f}} = \frac{1}{\sqrt{f_1}} + \frac{1}{\sqrt{f_2}} + \frac{1}{\sqrt{f_3}}$$

**Answer: C**

## Solution:

### Given:

A stretched wire has a **fundamental frequency**  $f$ .

When it is divided into **3 segments**, their fundamental frequencies are  $f_1$ ,  $f_2$ , and  $f_3$ .

Tension  $T$  is **constant**.

We need to find the relation among  $f$ ,  $f_1$ ,  $f_2$ ,  $f_3$ .

### Step 1: Formula for fundamental frequency of a stretched string

For a string under tension  $T$ , length  $L$ , and linear density  $\mu$ :

$$f = \frac{1}{2L} \sqrt{\frac{T}{\mu}}$$

### Step 2: Frequencies of the three parts

Suppose the original length is divided into three parts of lengths  $L_1$ ,  $L_2$ ,  $L_3$ , so that

$$L = L_1 + L_2 + L_3$$

Then their fundamental frequencies are

$$f_1 = \frac{1}{2L_1} \sqrt{\frac{T}{\mu}}, \quad f_2 = \frac{1}{2L_2} \sqrt{\frac{T}{\mu}}, \quad f_3 = \frac{1}{2L_3} \sqrt{\frac{T}{\mu}}$$

### Step 3: Express $L_i$ in terms of $f_i$

From the above,

$$L_i = \frac{1}{2f_i} \sqrt{\frac{T}{\mu}}$$

and

$$L = \frac{1}{2f} \sqrt{\frac{T}{\mu}}$$

### Step 4: Use $L = L_1 + L_2 + L_3$

$$\frac{1}{2f} \sqrt{\frac{T}{\mu}} = \frac{1}{2} \sqrt{\frac{T}{\mu}} \left( \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3} \right)$$

Cancel the common factor  $\frac{1}{2} \sqrt{\frac{T}{\mu}}$ :

$$\boxed{\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}}$$

 **Correct Option: (C)**

$$\boxed{\frac{1}{f} = \frac{1}{f_1} + \frac{1}{f_2} + \frac{1}{f_3}}$$

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## Question8

A steel wire of length 81 cm has a mass of  $5 \times 10^{-3}$  kg.

If the wire is under a tension of 50 N , then the speed of transverse waves on the wire is

**AP EAPCET 2025 - 22nd May Morning Shift**

**Options:**

A.

$$100 \text{ ms}^{-1}$$

B.

$$105 \text{ ms}^{-1}$$

C.

$$90 \text{ ms}^{-1}$$

D.

$$60 \text{ ms}^{-1}$$

**Answer: C**

**Solution:**

Mass per unit length,

$$\mu = \frac{M}{l} = \frac{5 \times 10^{-3}}{0.81}$$

$\therefore$  Speed of transverse wave

$$\begin{aligned} v &= \sqrt{\frac{T}{\mu}} = \sqrt{\frac{50}{\frac{5 \times 10^{-3}}{0.81}}} \\ &= \sqrt{\frac{50 \times 0.81}{5 \times 10^{-3}}} = 90 \text{ m/s} \end{aligned}$$

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## Question9

The speed of a stationary wave represented by the equation

$$y = 0.7 \sin\left(\frac{7\pi}{4}x\right) \cos(350\pi t) \text{ is}$$

(In the given equation  $x$  and  $y$  are in metre and  $t$  is in second)

**AP EAPCET 2025 - 21st May Evening Shift**

**Options:**

A.

$$100 \text{ ms}^{-1}$$

B.

$$150 \text{ ms}^{-1}$$

C.

$$160 \text{ ms}^{-1}$$

D.

$$200 \text{ ms}^{-1}$$

**Answer: D**

**Solution:**

$$y = 0.7 \sin\left(\frac{7\pi}{4}x\right) \cos(350\pi t)$$

Comparing with equation of stationary wave,

$$y = 2A \sin(kx) \cos \omega t$$

$$\text{We get, } k = \frac{7\pi}{4}, \omega = 350\pi$$

$\therefore$  Speed of stationary wave,

$$v = \frac{\omega}{k} = \frac{350\pi}{\frac{7\pi}{4}} = 200 \text{ m/s}$$

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## Question10

Two sound waves of wavelengths 99 cm and 100 cm produce 10 beats in a time of  $t$  seconds. If the speed of sound in air is  $330 \text{ ms}^{-1}$ , then the value of  $t$  in seconds is

AP EAPCET 2025 - 21st May Morning Shift

Options:

A.

12

B.

9

C.

6

D.

3

**Answer: D**

**Solution:**

$$f_1 = \frac{v}{\lambda_1} = \frac{330}{0.99} = 333.33 \text{ Hz}$$

$$f_2 = \frac{v}{\lambda_2} = \frac{330}{1} = 330 \text{ Hz}$$

$$\therefore f_{\text{beat}} = f_1 - f_2 = 333.33 - 330 = 3.33 \text{ Hz}$$

$$\therefore t = \frac{\text{Total number of beat}}{f_{\text{beat}}} = \frac{10}{3.33} = 3 \text{ s}$$

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## Question11

**The frequency of fifth harmonic of a closed organ pipe is equal to the frequency of third harmonic of an open organ pipe. If the length of the open pipe is 72 cm , then length of the closed organ pipe is**

### **AP EAPCET 2024 - 23th May Morning Shift**

**Options:**

A. 60 cm

B. 45 cm

C. 30 cm

D. 75 cm

**Answer: A**

### **Solution:**

Here's how to find the closed-pipe length  $L_c$ :

For a closed pipe (one end closed), only odd harmonics appear:

$$f_m = \frac{mv}{4L_c} \quad (m = 1, 3, 5, \dots)$$

For an open pipe (both ends open), all harmonics appear:

$$f_n = \frac{nv}{2L_o} \quad (n = 1, 2, 3, \dots)$$

Equate the 5th harmonic of the closed pipe to the 3rd harmonic of the open pipe:

$$\frac{5v}{4L_c} = \frac{3v}{2L_o}$$

Solve for  $L_c$ :

$$\frac{5}{4L_c} = \frac{3}{2L_o} \implies L_c = \frac{5}{6} L_o = \frac{5}{6} \times 72 \text{ cm} = 60 \text{ cm}$$

Answer: **60 cm** (Option A).

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## **Question12**

**The fundamental frequency of an open pipe is 100 hz If the bottom end of the pipe is closed and 1/3 rd of the pipe is filled with water, then the fundamental frequency of the pipe is**

# AP EAPCET 2024 - 22th May Evening Shift

## Options:

A. 200 Hz

B. 100 Hz

C. 75 Hz

D. 150 Hz

**Answer: C**

## Solution:

To determine the new fundamental frequency when changing the configuration of the pipe, follow these steps:

### Initial Conditions

**Length of the pipe (L):** The total length of the open pipe.

**Speed of sound (v):** A constant parameter in this context.

### Open Pipe Fundamental Frequency

For an open pipe, the fundamental frequency ( $f_0$ ) is given by:

$$f_0 = \frac{v}{2L}$$

Given that  $f_0 = 100$  Hz, we have:

$$100 = \frac{v}{2L} \Rightarrow v = 200L$$

### Closed Pipe Adjustment

When the pipe's bottom end is closed and  $\frac{1}{3}$  of the pipe is filled with water, the air column effectively resonating is:

$$L' = L - \frac{L}{3} = \frac{2L}{3}$$

### Closed Pipe Fundamental Frequency

For a closed pipe, the fundamental frequency ( $f_C$ ) is calculated using:

$$f_C = \frac{v}{4L'}$$

Substitute  $L'$  into the formula:

$$f_C = \frac{v}{4 \times \frac{2L}{3}} = \frac{3v}{8L}$$

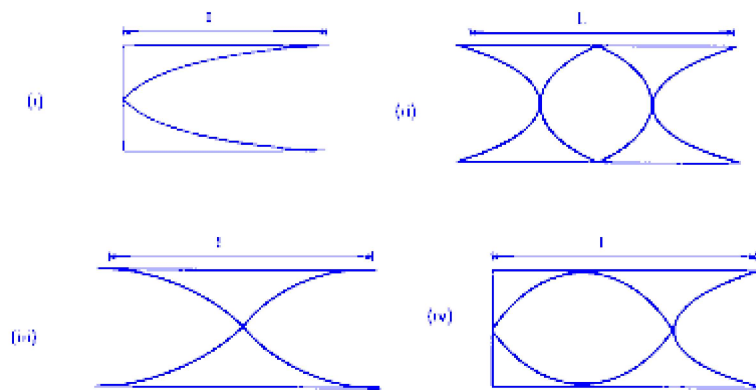
Plug in the value of  $v$  from the earlier expression:

$$f_C = \frac{3 \times 200L}{8 \times L} = \frac{600}{8} = 75 \text{ Hz}$$

Thus, the new fundamental frequency when the pipe is closed and partly filled with water is **75 Hz**.

## Question13

The vibrations of four air columns are shown below. The ratio of frequencies is



**AP EAPCET 2024 - 22th May Morning Shift**

**Options:**

A. 1 : 2 : 3 : 4

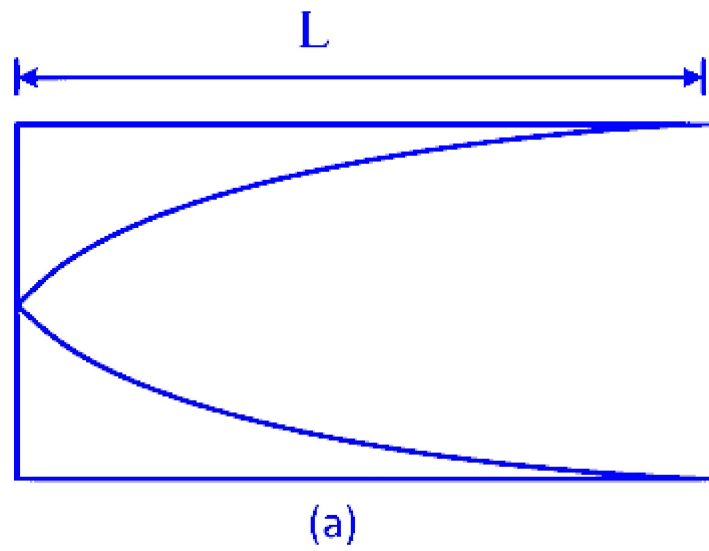
B. 1 : 3 : 2 : 4

C. 1 : 4 : 3 : 2

D. 1 : 4 : 2 : 3

**Answer: D**

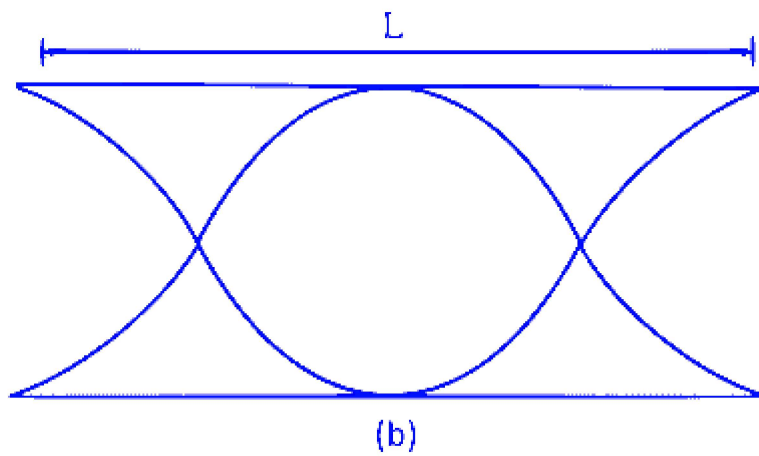
**Solution:**



The pipe is closed  $f = \frac{v}{\lambda}$

Here,  $L = \frac{\lambda}{4}$

$$\therefore f_a = \frac{v}{4L}$$

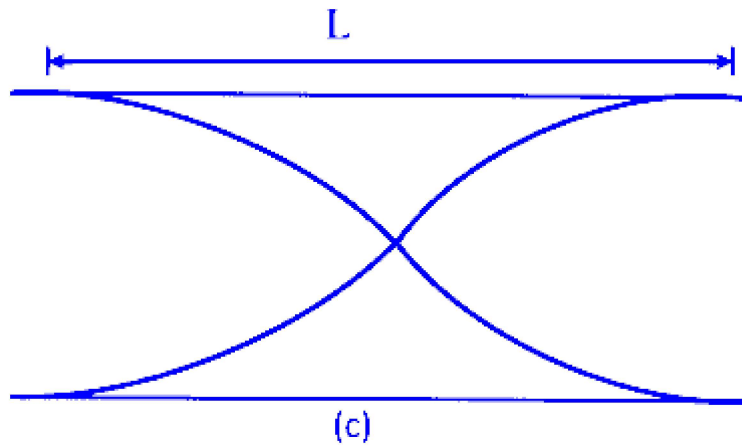


The pipe is open

So,  $f = \frac{v}{\lambda}$

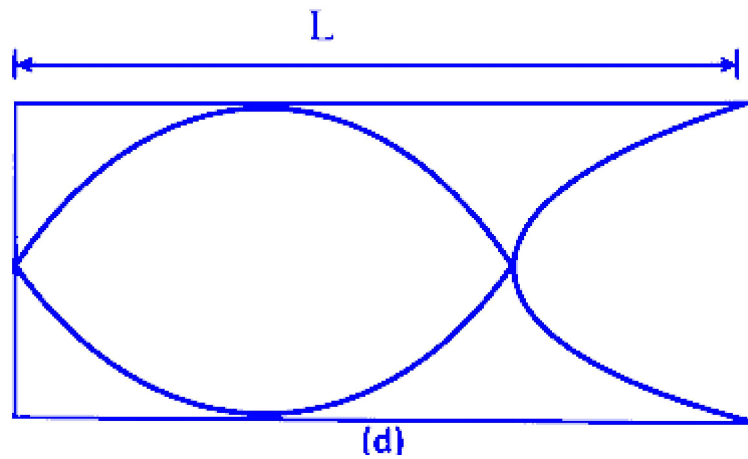
Here,  $L = \lambda$

$$f_b = \frac{v}{L}$$



The pipe is open,  $f = \frac{v}{\lambda}$

Here,  $L = \frac{\lambda}{2}, f_c = \frac{v}{2L}$



The pipe is closed

So,  $f = \frac{v}{\lambda}$

Here,  $L = \frac{3\lambda}{4} \Rightarrow f_d = \frac{3v}{4L}$

Now,  $f_a : f_b : f_c : f_d$

$\frac{v}{4L} : \frac{v}{L} : \frac{v}{2L} : \frac{3v}{4L} = 1 : 4 : 2 : 3$

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## Question14

**When a wave enters into a rarer medium from a denser medium, the property of the wave which remains constant is**

**AP EAPCET 2024 - 21th May Evening Shift**

**Options:**

- A. wavelength
- B. frequency
- C. velocity
- D. amplitude

**Answer: B**

### **Solution:**

When a wave transitions from a denser medium to a rarer medium, the property that remains constant is the **frequency**.

Frequency is determined by the source of the wave and remains unchanged when the wave passes from one medium to another.

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## **Question15**

**A car sounding a horn of frequency 1000 Hz passes a stationary observer. The ratio of frequencies of the horn noted by the observer before and after passing of the car is 11 : 9. The speed of car is (speed of sound  $v = 340 \text{ ms}^{-1}$  )**

### **AP EAPCET 2024 - 21th May Morning Shift**

**Options:**

- A.  $34 \text{ ms}^{-1}$
- B.  $17 \text{ ms}^{-1}$
- C.  $170 \text{ ms}^{-1}$
- D.  $340 \text{ ms}^{-1}$

**Answer: A**

### **Solution:**

Let's define:

$f = 1000 \text{ Hz}$ : the original frequency of the horn.

$v = 340 \text{ m/s}$ : the speed of sound.

$v_c$ : the speed of the car.

## Apparent Frequency Formulas

**Approaching Observer:**

$$f_1 = f \left( \frac{v}{v-v_c} \right)$$

**Receding from Observer:**

$$f_2 = f \left( \frac{v}{v+v_c} \right)$$

## Frequency Ratio Equation

Given  $\frac{f_1}{f_2} = \frac{11}{9}$ , we equate and solve:

$$\frac{f \left( \frac{v}{v-v_c} \right)}{f \left( \frac{v}{v+v_c} \right)} = \frac{v+v_c}{v-v_c} = \frac{11}{9}$$

## Solving for $v_c$

Starting from:

$$\frac{11}{9} = \frac{340+v_c}{340-v_c}$$

Cross-multiplying gives:

$$11(340 - v_c) = 9(340 + v_c)$$

Simplifying:

$$3740 - 11v_c = 3060 + 9v_c$$

$$680 = 20v_c$$

$$v_c = \frac{680}{20} = 34 \text{ m/s}$$

Thus, the speed of the car is 34 m/s.

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## Question16

**A pipe with 30 cm length is open at both ends. Which harmonic mode of the pipe resonates with the 1.65 kHz source? (Velocity of sound in air =  $330 \text{ ms}^{-1}$  )**

**Options:**

A. 2

B. 3

C. 3.5

D. 2.5

**Answer: B**

**Solution:**

Given:

Frequency,  $v_n = 1.65 \text{ kHz} = 1650 \text{ Hz}$

Velocity of sound,  $v = 330 \text{ m/s}$

Length of pipe,  $L = 30 \text{ cm} = 0.3 \text{ m}$

The frequency of the  $n$ th harmonic in an open pipe is calculated using the formula:

$$v_n = \frac{nv}{2L}$$

Substitute the given values:

$$1650 = \frac{n \times 330}{2 \times 0.3}$$

Solving for  $n$ :

$$1650 = \frac{n \times 330}{0.6}$$

$$1650 \times 0.6 = n \times 330$$

$$990 = n \times 330$$

$$n = \frac{990}{330} = 3$$

Thus, the pipe resonates at the 3rd harmonic.

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## Question17

**If the frequency of a wave is increased by 25%, then the change in its wavelength is (medium not changed)**

**AP EAPCET 2024 - 20th May Morning Shift**

**Options:**

A. 20% increase

B. 20% decrease

C. 25% increase

D. 25% decrease

**Answer: B**

### **Solution:**

When the frequency of a wave is increased by 25%, the wavelength changes as follows (assuming the medium remains unchanged).

The speed of the wave remains the same because the medium does not change:

$$v = \text{constant}$$

Therefore, according to the wave equation:

$$f_1 \lambda_1 = f_2 \lambda_2$$

If the initial frequency  $f_1$  is increased by 25%, the new frequency  $f_2$  becomes:

$$f_2 = 1.25 \times f_1$$

Substituting this into the wave equation:

$$f_1 \lambda_1 = (1.25 \times f_1) \lambda_2$$

Solving for the new wavelength  $\lambda_2$ :

$$\lambda_2 = \frac{\lambda_1}{1.25} = 0.8 \lambda_1$$

The change in wavelength  $\Delta \lambda$  is calculated as:

$$\Delta \lambda = \frac{\lambda_2 - \lambda_1}{\lambda_1} \times 100 = \frac{0.8 \lambda_1 - \lambda_1}{\lambda_1} \times 100$$

Simplifying this gives:

$$\Delta \lambda = \frac{-0.2 \lambda_1}{\lambda_1} \times 100 = -20\%$$

Thus, the wavelength decreases by 20%.

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## **Question 18**

**The speed of a wave on a string is  $150 \text{ ms}^{-1}$  when the tension is 120 N . The percentage increase in the tension in order to raise the wave speed by 20% is**

# AP EAPCET 2024 - 19th May Evening Shift

**Options:**

A. 44

B. 40

C. 22

D. 20

**Answer: A**

**Solution:**

To determine the percentage increase in tension required to raise the wave speed by 20%, we start by examining the given relationship between wave speed and tension.

## Step-by-Step Solution

**Calculate the New Wave Speed:**

Given that the original wave speed is  $v_1$  and we need to increase it by 20%, the new speed  $v_2$  is:

$$v_2 = v_1 + 20\% \text{ of } v_1 = v_1 + \frac{v_1}{5} = \frac{6v_1}{5}$$

**Relate Wave Speed to Tension:**

The speed of a wave on a string is given by:

$$v = \sqrt{\frac{T}{m}}$$

This implies:

$$\frac{v_2}{v_1} = \sqrt{\frac{T_2}{T_1}}$$

Squaring both sides gives:

$$\frac{T_2}{T_1} = \left(\frac{v_2}{v_1}\right)^2$$

**Substitute  $v_2$  and  $v_1$ :**

$$\frac{T_2}{T_1} = \left(\frac{\frac{6v_1}{5}}{v_1}\right)^2 = \left(\frac{6}{5}\right)^2 = \frac{36}{25}$$

**Find the Required Increase in Tension:**

$$\frac{T_2 - T_1}{T_1} = \frac{T_2}{T_1} - 1 = \frac{36}{25} - 1 = \frac{11}{25}$$

### Convert to Percentage:

$$\frac{T_2 - T_1}{T_1} \times 100\% = \frac{11}{25} \times 100\% = 44\%$$

Thus, the tension needs to be increased by **44%** to achieve a 20% increase in wave speed.

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## Question19

Two stretched strings  $A$  and  $B$  when vibrated together produce 4 beats per second. If the tension applied to the string  $A$  increased. The number of beats produced per second is increased to 7. If the frequency of string  $B$  is 480 Hz initially, the frequency of string  $A$  is

### AP EAPCET 2024 - 18th May Morning Shift

#### Options:

A. 473 Hz

B. 476 Hz

C. 484 Hz

D. 487 Hz

**Answer: C**

#### Solution:

To determine the frequency of string  $A$ , we start with the following information:

Initial beat frequency is 4 beats per second.

Frequency of string  $B$ ,  $f_B = 480$  Hz.

The relation for beat frequency:  $|f_A - f_B| = 4$ .

Given that the tension on string  $A$  is increased, the frequency of  $A$  will increase, leading to an increase in the beat frequency. The new beat frequency becomes 7 beats per second.

The vibration frequency of a string is determined by the formula:

$$f = \frac{n}{2L} \cdot \sqrt{\frac{T}{\mu}}$$

Where:

$n$  is the mode of vibration.

$L$  is the length of the string.

$T$  is the tension in the string.

$\mu$  is the linear mass density.

Since the tension in string  $A$  is increased, we know the frequency of  $A$  increases. Consequently, the following must be true for the beat frequency to increase:

$$f_A - f_B = 4 \quad \text{or} \quad f_B - f_A = 4$$

However, since the frequency of  $A$  increases and the beat frequency becomes 7, it implies that:

$$f_A - f_B = 7$$

Given:

Initially,  $f_A - 480 = 4$ , thus  $f_A = 484$  Hz.

Therefore, as  $f_A$  increases past 480 Hz and still satisfies the condition where the beat frequency becomes 7, this confirms that initially:

$$f_A = 484 \text{ Hz}$$

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## Question20

**Two cars are moving towards each other at the speed of  $50 \text{ ms}^{-1}$ . If one of the cars blows a horn at a frequency of 250 Hz, the wave length of the sound perceived by the driver of the other car is**

**(Speed of sound in air =  $350 \text{ ms}^{-1}$ )**

### AP EAPCET 2022 - 5th July Morning Shift

**Options:**

A. 18.7 cm

B. 105 cm

C. 75 cm

D. 10.5 cm

**Answer: B**

## Solution:

Here, both cars are moving towards each other, hence one car is like an observer and another car is like a source.

$$\therefore v_s = 50 \text{ m/s} \Rightarrow v_0 = -50 \text{ m/s}$$

Frequency of horn blown by the source car,

$$v = 250 \text{ Hz}$$

According to Doppler's effect, frequency heard by another car  $v'$  is given as

$$v' = \frac{v-v_0}{v-v_s} \times v = \frac{350-(-50)}{350-50} \times 250$$

$$\therefore (v = \text{speed of sound} = 350 \text{ m/s})$$

$$= \frac{400 \times 250}{300} \Rightarrow v' = \frac{100}{3} \text{ Hz}$$

Wavelength of heard sound,

$$\begin{aligned} \lambda' &= \frac{v}{v'} \\ &= \frac{350}{1000/3} = \frac{1050}{1000} = 1.05 \text{ m} = 105 \text{ cm} \end{aligned}$$

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## Question21

Speed of sound in air near room temperature is approximately

**AP EAPCET 2022 - 4th July Evening Shift**

**Options:**

A.  $3.4 \times 10^2 \text{ ms}^{-1}$

B.  $34 \text{ ms}^{-1}$

C.  $34 \text{ kms}^{-1}$

D.  $3.4 \text{ km s}^{-1}$

**Answer: A**

**Solution:**

The speed of sound in air at room temperature is approximately 340 m/s or  $3.4 \times 10^2$  m/s.

---

## Question22

**A body is suspended from a string of length 1 m and mass 2 g. The mass of the body to produce a fundamental mode of 100 Hz frequency in the string is (Acceleration due to gravity =  $10 \text{ ms}^{-2}$ )**

### AP EAPCET 2022 - 4th July Morning Shift

**Options:**

- A. 80 g
- B. 4 kg
- C. 400 g
- D. 8 kg

**Answer: D**

**Solution:**

Given, mass of string,  $m = 2 \text{ g}$

$$= 2 \times 10^{-3} \text{ kg}$$

length of the string,  $l = 1 \text{ m}$

$\therefore$  Mass per unit length

$$\mu = \frac{m}{l} = 2 \times 10^{-3} \text{ kg/m}$$

Tension in the string,

$$T = \text{mass of body} \times g$$

$$T = m'g$$

Frequency of transverse wave is given as

$$f = \frac{1}{2l} \sqrt{\frac{T}{\mu}}$$

$$\Rightarrow 100 = \frac{1}{2 \times 1} \sqrt{\frac{m'g}{2 \times 10^{-3}}} \quad (\because \text{Given, } f = 100 \text{ Hz})$$

$$\Rightarrow (200)^2 = \frac{m'g}{2 \times 10^{-3}}$$

$$\Rightarrow m' = \frac{4 \times 10^4 \times 2 \times 10^{-3}}{g} = \frac{80}{10} = 8 \text{ kg}$$

## Question23

**Two waves are represented by**

$x_1 = A \sin \left( \omega t + \frac{\pi}{6} \right)$  and  $x_2 = A \cos \omega t$ . **Then, the phase difference between them is**

**AP EAPCET 2021 - 20th August Evening Shift**

**Options:**

A.  $\frac{\pi}{6}$

B.  $\frac{\pi}{2}$

C.  $\frac{\pi}{3}$

D.  $\pi$

**Answer: C**

**Solution:**

We are given two waves:

$$x_1 = A \sin \left( \omega t + \frac{\pi}{6} \right) \text{ and } x_2 = A \cos \omega t$$

**Step 1: Find the phase of each wave**

For the first wave,  $x_1$ , the phase is the angle inside the sine function: Phase of  $x_1(\phi_1) = \frac{\pi}{6}$

For the second wave,  $x_2 = A \cos \omega t$ , we can write cosine as sine:  $\cos \omega t = \sin \left( \omega t + \frac{\pi}{2} \right)$  So,  
 $x_2 = A \sin \left( \omega t + \frac{\pi}{2} \right)$  The phase of  $x_2$  is: Phase of  $x_2(\phi_2) = \frac{\pi}{2}$

**Step 2: Find the phase difference**

The phase difference between the waves is:  $\Delta\phi = \phi_2 - \phi_1$

Substitute the values:  $\Delta\phi = \frac{\pi}{2} - \frac{\pi}{6}$

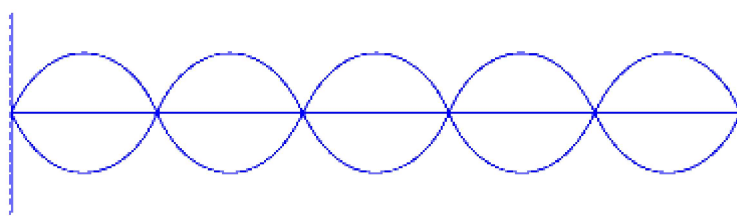
Find a common denominator and subtract:  $= \frac{3\pi}{6} - \frac{\pi}{6} = \frac{2\pi}{6} = \frac{\pi}{3}$

So, the phase difference between the waves is  $\frac{\pi}{3}$ .

---

## Question 24

**A string fixed at both ends vibrates in 5 loops as shown in the figure. The total number of nodes and anti-nodes respectively are**



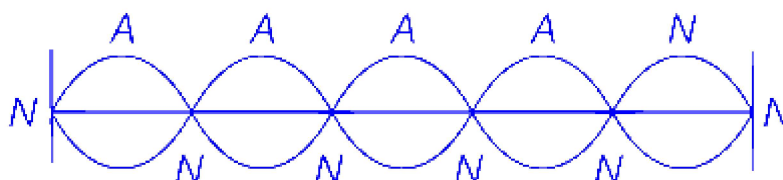
### AP EAPCET 2021 - 20th August Morning Shift

**Options:**

- A. 6 and 5
- B. 6 and 10
- C. 2 and 5
- D. 10 and 6

**Answer: A**

**Solution:**



From above wave diagram,

Number of node,  $N = 6$

and number of anti-node,  $A = 5$

---

## Question 25

Match the following.

|     | Column I                               |    | Column II                                               |
|-----|----------------------------------------|----|---------------------------------------------------------|
| (A) | Transverse wave through a steel rod    | 1. | $\sqrt{B + \left(\frac{4}{3}\right) \frac{\eta}{\rho}}$ |
| (B) | Longitudinal waves in Earth's crust    | 2. | $\sqrt{\frac{\eta}{\rho}}$                              |
| (C) | Longitudinal waves through a steel rod | 3. | $\sqrt{\frac{2\pi T}{\rho \lambda}}$                    |
| (D) | Ripples                                | 4. | $\sqrt{\frac{\lambda}{\rho}}$                           |

### AP EAPCET 2021 - 19th August Evening Shift

Options:

A. A - 2, B - 1, C - 4, D - 3

B. A - 1, B - 3, C - 4, D - 2

C. A - 3, B - 4, C - 1, D - 2

D. A - 2, B - 4, C - 1, D - 3

**Answer: A**

**Solution:**

Speed of transverse wave,  $v = \sqrt{\eta/\rho}$

where,  $\eta$  = modulus of elasticity  $\rho$  = density

$\therefore (A) \rightarrow (2)$

Speed of longitudinal wave in earth crust

$$v = \sqrt{B + \frac{4}{3} \left( \frac{\eta}{\rho} \right)}$$

$\therefore$  (B)  $\rightarrow$  (1)

Speed of longitudinal wave,  $v = \sqrt{\lambda/\rho}$  where,  $\lambda$  = wavelength of wave.

$\therefore$  (C)  $\rightarrow$  (4)

$$\text{Ripples speed, } v = \sqrt{\frac{2\pi T}{\lambda \rho}}$$

where,  $T$  = surface tension coefficient.

$\therefore$  (D)  $\rightarrow$  (3)

---

## Question 26

The sources of sound A and B produce a wave of 350 Hz in same phase. A particle P is vibrating under an influence of these two waves. If the amplitudes at P produced by the two waves are 0.3 mm and 0.4 mm, the resultant amplitude of the point P will be, when  $AP - BP = 25$  cm and the velocity of sound is  $350 \text{ ms}^{-1}$

### AP EAPCET 2021 - 19th August Morning Shift

**Options:**

A. 0.7 mm

B. 0.1 mm

C. 0.2 mm

D. 0.5 mm

**Answer: D**

**Solution:**

Given, frequency of wave,  $f = 350 \text{ Hz}$

Amplitudes of wave  $a_1, a_2$  are 0.3 mm and 0.4 mm, respectively.

Path difference ( $AP - BP = \Delta x$ ) = 25 cm

Speed of sound,  $v = 350 \text{ ms}^{-1}$

As, we know that,

$$\Delta\phi = \frac{2\pi}{\lambda} \Delta x$$

$$\text{and (wavelength) } \lambda = \frac{(\text{speed}) v}{(\text{frequency}) f}$$

$$= \frac{350}{350} = 1 \text{ m} = 100 \text{ cm}$$

$$\Delta\phi = \frac{2\pi}{100} \times 25 = \pi/2$$

As, resultant amplitude,

$$a_R = \sqrt{a_1^2 + a_2^2 + 2a_1a_2 \cos \phi}$$

$$\Rightarrow a_R = \sqrt{0.3^2 + 0.4^2 + 2 \times 0.3 \times 0.4 \cos \pi/2}$$

$$= \sqrt{9/100 + 16/100}$$

$$= \sqrt{25/100}$$

$$= 5/10 = 0.5 \text{ mm}$$

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